Chemistry

Unit 1

Practice Exam 2

This sample exam paper has been prepared as part of Pearson suite of resources for the Year 11, Unit 1, ATAR Chemistry Course prescribed by the Western Australian School Curriculum and Standards Authority.

Time allowed

Reading time: 10 minutes Working time: 90 minutes

Materials required

An approved non-programmable calculator

Chemistry Data Sheet. This may be downloaded form the WACE website.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section one: Multiple-choice	13	13	25	13	25
Section two: Short answer	8	8	35	40	40
Section three: Extended answer	3	3	30	37	35
				Total	**100**

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Section one: Multiple-choice 25% (13 marks)

This section has **13** questions. Answer **all** questions by circling the correct option.

Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.

Suggested working time: 25 minutes

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1 An ion that contains 11 protons, 12 neutrons and 10 electrons will have a charge and mass number corresponding to:

	Mass number	Charge
a	11	–1
b	11	+1
c	23	–1
d	23	+1

2 An element has atomic number 3. In the periodic table, it would be found in

a period 1.

b group 1.

c period 3.

d group 3.

3 Elements are placed into groups on the periodic table according to each element's

a mass number.

b atomic number.

c number of occupied electron shells.

d number of electrons in the outer shell.

The following information refers to questions 4 and 5.

The atomic number, mass number and electron configuration of four particles, W, X, Y and Z, are given below.

	Atomic number	Mass number	Electron configuration
W	17	37	2,8,8
X	19	39	2,8,8
Y	20	40	2,8,8,2
Z	19	40	2,8,8,1

4 Which one of the following alternatives lists particles that are isotopes of the same element?

a W and X only

b Y and Z only

c X and Z only

d W, X and Y only

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5 Which one of the following statements about particles W, X, Y and Z is correct?

a W is a noble gas.

b X is a positively charged ion.

c Y is in group 4 of the periodic table.

d Z is a negatively charged ion.

6 Which one of the following alternatives lists particles that do **not** have the same electron configuration?

a Al^{3+} and Ca^{2+}

b Cl^{-} and Ca^{2+}

c Cl^{-} and Ar

d Ar and Ca^{2+}

7 Identify the substance that does **not** contain freely moving charged particles when in solid form.

a graphite

b copper

c silicon

d aluminium

8 Which one of the following statements about ionic bonding is **not** correct?

a An ionic solid contains both cations and anions in fixed positions.

b Ionic bonding involves the sharing of electrons between two different atoms.

c Compounds that contain ionic bonding tend to have high melting temperatures.

d Compounds that contain ionic bonding are able to conduct electricity when molten.

9 Atom P has 5 electrons in its outer shell and atom Q has 7 electrons in its outer shell. Identify the most likely empirical formula for a compound consisting of atoms P and Q.

a P_2Q

b PQ_2

c P_3Q

d PQ_3

10 Identify the compound with the highest percentage by mass of nitrogen.

a NH_3

b N_2H_4

c HNO_3

d NH_4NO_3

11 The hydrocarbon with the formula C_3H_8 is

a propane.

b 1-propane.

c 1-propene.

d propene.

12 When liquid water is heated, it forms gaseous water or steam. Therefore, for the reaction $H_2O(l) \rightarrow H_2O(g)$

- **a** ΔH is positive and the reaction is endothermic.
- **b** ΔH is negative and the reaction is endothermic.
- **c** ΔH is positive and the reaction is exothermic.
- **d** ΔH is negative and the reaction is exothermic.

13 The correct unit for the molar mass of an element is

- **a** gram.
- **b** mol.
- **c** gram mol^{-1}.
- **d** mol $gram^{-1}$.

End of section one

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Section two: **Short answer** 40% (40 marks)

This section has **8** questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 35 minutes

Question 14 **(5 marks)**

a Complete the table below by writing the formula of each of the compounds listed. (3 marks)

Name of compound	Formula of compound
Magnesium chloride	
Aluminium sulfate	
Tin(IV) oxide	

b Complete the table below by writing the name of each of the compounds listed. (2 marks)

Formula of compound	Name of compound
$Fe(NO_3)_3$	
NH_4Cl	

Question 15 **(6 marks)**

Give the chemical formula of each of the following:

a The element with one more proton than iron. (1 mark)

b The element with the highest electronegativity in group 1. (1 mark)

c The ion formed by the most reactive element in group 17. (1 mark)

d The element with only 3 occupied electron shells and 2 valence electrons. (1 mark)

e One of the actinides. (1 mark)

f The noble gas in period 2. (1 mark)

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Question 16 **(7 marks)**

For each of the following, determine which represents the larger quantity and give an explanation your choice.

a The atomic radius of the potassium atom or the atomic radius of the rubidium atom (2 marks)

b The first ionisation energy of sulfur or the first ionisation energy of magnesium (3 marks)

c The electrical conductivity of liquid propane and the electrical conductivity of liquid potassium chloride (2 marks)

Question 17 **(4 marks)**

Nanotechnology is an emerging area of science.

a Circle the quantity below that is equivalent to one nanometer. (1 mark)

10^{-3} m 10^{-6} m 10^{-9} m 10^{-12} m 10^{-18} m 10^{-23} m

b Explain why nanoparticles behave differently from much larger particles of the same substance. (2 marks)

c Outline one current application of nanotechnology. (1 mark)

Question 18 **(6 marks)**

Graphite and diamond are allotropes of carbon, although they have significantly different properties.

a What is meant by the term 'allotrope'? (1 mark)

b Describe the structures of graphite and diamond with reference to the different ways carbon is bonded in each substance. (4 marks)

c Graphite is used as 'lead' in pencils. With reference to the bonding in the lattice, explain why graphite can be used this way. (1 mark)

Question 19 **(4 marks)**

Iridium has two naturally occurring isotopes. Their relative abundances and masses are shown in the table below.

Isotope	Relative isotopic mass	Percentage abundance
^{191}Ir	190.97	37.30%
^{193}Ir	192.97	62.70%

a What is the name of the instrument that is commonly used to experimentally obtain the abundances and relative isotopic masses data? (1 mark)

b Using the information in the table, determine the relative atomic mass of iridium. Show your working and give your answer to the appropriate number of significant figures. (2 marks)

c Atomic absorption spectroscopy is an analytical technique that can be used to identify metals such as iridium. Upon what does this analytical technique rely? (1 mark)

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Question 20 **(4 marks)**

An intake of vitamin C is essential in a well-balanced diet. Vitamin C occurs naturally in fruit and many vegetables. The relative molecular mass of vitamin C is 176.1.

a A particular brand of orange juice claims to contain 40.0 mg of vitamin C in a 100 mL serve. Calculate the amount, in mol, of vitamin C in the 100 mL serve of that orange juice. (1 mark)

b How many molecules of vitamin C are in 100 mL of that orange juice? (1 mark)

c Vitamin C contains 54.5% oxygen by mass. How many atoms of oxygen are in one molecule of vitamin C? (2 marks)

Question 21 **(4 marks)**

A compound contains 12.8% carbon, 2.13% hydrogen and the rest is bromine.

a Use this information to determine the empirical formula of the compound. (3 marks)

b Given that the relative molecular mass of the compound is 188, determine its molecular formula. (1 mark)

End of section two

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Section three: Extended answer 35% (37 marks)

This section has **3** questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 30 minutes

Question 22 **(11 marks)**

Consider the compound with the molecular formula C_2H_6.

a To what group of organic compounds does it belong? (1 mark)

b Name the compound. (1 mark)

This compound reacts readily with chlorine gas in the presence of ultraviolet light.

c Write an equation for the reaction between 1 mole of C_2H_6 and 1 mole of chlorine gas in the presence of ultra-violet light. (1 mark)

d **i** Identify the bonds that are broken in this reaction. (1 mark)

ii Does bond breaking absorb or release energy? (1 mark)

e **i** Identify the bonds that are formed in this reaction. (1 mark)

ii Does bond forming absorb or release energy? (1 mark)

f Given that energy is released in the overall reaction, deduce whether the bonds formed are overall stronger or weaker than the bonds broken. (1 mark)

g **i** Explain what is meant by the phrase 'heat of reaction'. (1 mark)

ii Will the heat of reaction between C_2H_6 and chlorine have a positive or negative value? (1 mark)

iii Is this reaction classified as exothermic or endothermic? (1 mark)

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Question 23 **(14 marks)**

Research into energy sources is increasing as climate debate intensifies.

a A power station uses about 36 000 tonnes of coal each day (1 tonne = 10^6 g) which contains 23.0% carbon by mass. The equation for the reaction that releases energy is:

$C(s) + O_2(g) \rightarrow CO_2(g)$

i Calculate the amount, in mol, of carbon used each day. (2 marks)

ii Calculate the mass, in grams, of carbon dioxide produced each day. (2 marks)

iii Calculate the mass of coal, in kg, that would be used when 10.0 kg of oxygen gas reacts. (3 marks)

b Biogas, a biofuel, is one source of renewable energy that is showing promise. The main constituent of biogas is methane.

i Write a balanced equation for the complete combustion of methane. (2 marks)

ii When 1.00 mol of methane is completely combusted, 890 kJ energy is released. How much energy is released when 1.00 g of methane burns? (2 marks)

iii Give the name of one other biofuel. (1 mark)

c Hydrogen gas can be used as a fuel in a hydrogen powered vehicle. The equation for the reaction is:

$2H_2(g) + O_2(g) \rightarrow 2H_2O(g) \quad \Delta H = -572 \text{ kJ mol}^{-1}$

Calculate the mass, in gram, of hydrogen required to produce 3.00×10^3 kJ of energy. (2 marks)

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Question 24 **(12 marks)**

a Draw the structural formula and give the systematic name for an organic compound that fits each of the following descriptions. (6 marks)

Description	Structural formula	Systematic name
A branched alkane with a total of 5 carbon atoms		
A compound that has the molecular formula C_4H_9Cl		
A compound with an empirical formula CH_2		

b What is meant by the term 'unsaturated' when referring to hydrocarbons? (1 mark)

c **i** Give the name or formula of a compound that could be used to test whether a hydrocarbon was saturated or unsaturated. (1 mark)

ii Describe what you would expect to see if the compound identified in (i) was added to a saturated and to an unsaturated compound. (2 marks)

- Saturated compound
- Unsaturated compound

d Write an equation for the reaction between an unsaturated hydrocarbon with 3 carbon atoms and chlorine gas. Use structural or semi-structural formulas for all carbon compounds in the equation. (2 marks)

End of questions

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